MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – TEST (SET - A)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM:** 25

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. At some time of the day, the length of the shadow of a tower is equal to its height. Then, the Sun's altitude at that time is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Since shadow length equals height, $\tan\theta = \dfrac{h}{h} = 1 = \tan 45° \implies \theta = \boxed{45°}$.

Q2. If a vertical pole of length 7.5 m casts a shadow 5 m long on the ground and at the same time, a tower casts a shadow 24 m long, then the height of the tower is: [1]

(A) 20 m (B) 40 m (C) 60 m (D) 36 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: By proportional shadow lengths at the same time of day: $\dfrac{H}{24} = \dfrac{7.5}{5} = \dfrac{3}{2} \implies H = 12 \times 3 = \boxed{36 \text{ m}}$.

Q3. From a point on the ground, which is 30 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be $60°$. The height (in metres) of the tower is: [1]

(A) $10\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) 60 m (D) 30 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: $h = 30\tan 60° = \boxed{30\sqrt{3} \text{ m}}$.

Q4. The length of the string of a kite flying 50 m above the ground with an angle of elevation of 60° is: [1]

(A) $\frac{100}{\sqrt{3}}$ m (B) $100\sqrt{3}$ m (C) 150 m (D) $\frac{50}{\sqrt{3}}$ m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\sin 60° = \frac{50}{L} \implies L = \frac{50}{\sqrt{3}/2} = \boxed{\frac{100}{\sqrt{3}} \text{ m}}$.

Q5. A ladder 14 m long leans against a wall. If the foot of the ladder is 7 m from the wall, then the angle of elevation of the top of the wall is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: $\cos\theta = \frac{7}{14} = \frac{1}{2} = \cos 60° \implies \theta = \boxed{60°}$.

SECTION B: Very Short Answer Questions

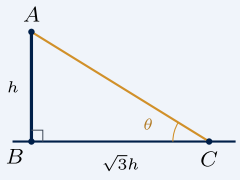
MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. The length of the shadow of a tower on the plane ground is $\sqrt{3}$ times the height of the tower. Find the angle of elevation of the Sun. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let tower $AB = h$ cast shadow $BC = \sqrt{3}h$, with Sun's elevation $\angle ACB = \theta$.

[0.5 Mark]



$$\tan\theta = \frac{h}{\sqrt{3}h} = \frac{1}{\sqrt{3}} = \tan 30° \implies \theta = \boxed{30°}$$

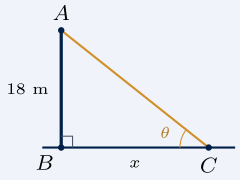
Final Sun elevation: [1.5 Marks]

Q7. Find the length of the shadow on the ground of a pole of height 18 m when the angle of elevation θ of the Sun is such that $\tan\theta = \frac{6}{7}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let pole $AB = 18$ m and shadow length $BC = x$ m.

[0.5 Mark]



$$\tan\theta = \frac{AB}{BC} = \frac{18}{x} = \frac{6}{7} \implies x = \frac{18 \times 7}{6} = \boxed{21 \text{ m}}$$

Shadow length calculation: [1.5 Marks]

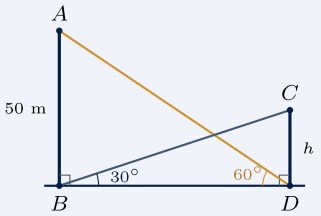
Q8. The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the tower is 50 m high, find the height of the building. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = 50$ m and building $CD = h$ m separated by horizontal distance $BD = x$ m, with $\angle ADB = 60°$ and $\angle CBD = 30°$. [0.5 Mark]

In right $\triangle ABD$: $\tan 60° = \dfrac{50}{x} \implies x = \dfrac{50}{\sqrt{3}}$ m. [1 Mark]

In right $\triangle CDB$: $\tan 30° = \dfrac{h}{x} \implies h = \dfrac{x}{\sqrt{3}}$. [1 Mark]



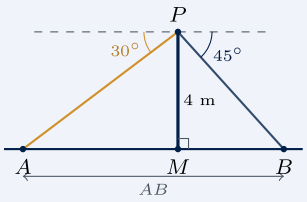
Substituting $x = \dfrac{50}{\sqrt{3}}$: $h = \dfrac{50/\sqrt{3}}{\sqrt{3}} = \dfrac{50}{3} = \boxed{16\dfrac{2}{3} \text{ m} \approx 16.67 \text{ m}}$. [0.5 Mark]

Q9. From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45° respectively. If the bridge is at a height of 4 m from the banks, find the width of the river. (Use $\sqrt{3} = 1.732$) **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let bridge point P be $PM = 4$ m above opposite river banks A (depression 30°) and B (depression 45°). [0.5 Mark]

In $\triangle PMA$: $AM = 4\cot 30° = 4\sqrt{3}$ m. In $\triangle PMB$: $MB = 4\cot 45° = 4$ m. [2 Marks]



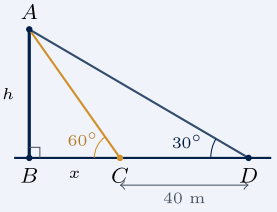
Width of the river $= AB = 4\sqrt{3} + 4 = 4(1.732 + 1) = 4 \times 2.732 = \boxed{10.928 \text{ m}}$. [0.5 Mark]

Q10. The shadow of a tower standing on level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60°. Find the height of the tower. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$ m, shadow at 60° be $BC = x$ m, and at 30° be $BD = (x + 40)$ m ($CD = 40$ m). [0.5 Mark]

$x = \dfrac{h}{\sqrt{3}}$ and $x + 40 = h\sqrt{3} \implies h\left(\sqrt{3} - \dfrac{1}{\sqrt{3}}\right) = 40 \implies \dfrac{2h}{\sqrt{3}} = 40$. [2 Marks]



$\implies h = 20\sqrt{3} = 20(1.732) = \boxed{20\sqrt{3} \text{ m} \approx 34.64 \text{ m}}$. [0.5 Mark]

Note: The source question paper header specifies MM: 25, while the printed section breakdown totals $5+4+9+5=23$ Marks.

Q11. A pole 6 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point P on the ground is $60°$ and the angle of depression of the point P from the top of the tower is $45°$. Find the height of the tower and the distance of point P from the foot of the tower. (Use $\sqrt{3} = 1.73$) **[5]**

OR

From the top of a building 60 m high, the angles of depression of the top and bottom of a vertical lamp post are observed to be $30°$ and $60°$ respectively.

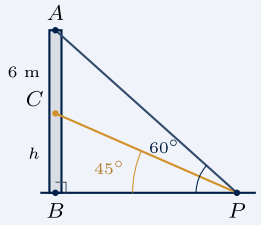
(i) Find the horizontal distance between the building and the lamp post.

(ii) Find the height of the lamp post and the distance between the tops of the building and the lamp post.

[5]

▶ MAIN QUESTION — STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let tower be $BC = h$ m topped by pole $AC = 6$ m. From P on ground, $\angle APB = 60°$, and depression of P from tower top C is $45° \implies \angle CPB = 45°$. [1 Mark]



In right $\triangle CBP$: $\tan 45° = \dfrac{h}{BP} \implies BP = h$. [1 Mark]

In right $\triangle ABP$: $\tan 60° = \dfrac{h+6}{h} \implies \sqrt{3}\,h = h + 6$. [1.5 Marks]

$h(\sqrt{3}-1) = 6 \implies h = \dfrac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1)$. With given $\sqrt{3} = 1.73$:

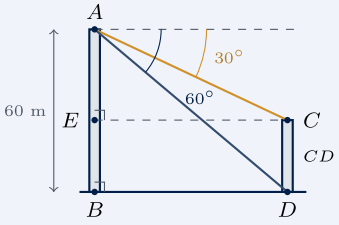
$$h = 3(1.73+1) = 3 \times 2.73 = 8.19 \text{ m}, \qquad BP = h = 8.19 \text{ m}$$

$$\therefore \text{ Height of tower } = \boxed{8.19 \text{ m}}, \qquad \text{Distance of point } P = \boxed{8.19 \text{ m}}$$

Rationalizing and substituting $\sqrt{3} = 1.73$: [1.5 Marks]

▶ OR CHOICE — STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let building be $AB = 60$ m and lamp post be $CD = h$ m. Draw $CE \perp AB$ ($BE = h$, $AE = 60 - h$). By alternate angles, $\angle ACE = 30°$ and $\angle ADB = 60°$. [1 Mark]



(i) In right $\triangle ABD$: $\tan 60° = \dfrac{60}{BD} \implies BD = \dfrac{60}{\sqrt{3}} = \boxed{20\sqrt{3} \text{ m} \approx 34.64 \text{ m}}$. [1.5 Marks]

(ii) In $\triangle AEC$: $CE = BD = 20\sqrt{3}$ m, so $AE = 20\sqrt{3}\tan 30° = 20$ m. [1 Mark]

Height of lamp post $= CD = BE = AB - AE = 60 - 20 = \boxed{40 \text{ m}}$. [0.5 Mark]

Distance between their tops AC (from right $\triangle AEC$): $\sin 30° = \dfrac{AE}{AC} \implies AC = \dfrac{20}{1/2} = \boxed{40 \text{ m}}$. [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★